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ABOUT ME

Third-year Statistical Data Science student at ISI Kolkata with a strong programming and problem-solving background. I build projects end to end, from the statistical or mathematical formulation through to working code, and pick up new tools and methods as each problem demands. Currently looking for **internships and programs** in Quantitative Finance, Machine Learning and Data Science.

EDUCATION

Indian Statistical Institute, Kolkata **2024–29**

Bachelor of Statistical Data Science (Hons.) | **76.5% (Distinction)** Jul 2024–May 2028
Integrated Master of Statistical Data Science *(upcoming)* Jul 2028–May 2029

SCHOLASTIC ACHIEVEMENTS

- **1st Runner-Up** in **QuantStorm 2026**, sponsored by **Optiver** and **Jane Street** [2026]
- **Rank 249 globally** and **AIR 43** at the **Prosperity 3** Challenge, **IMC Trading** [2025]
- **Rank 1** in category, **ISI Entrance Examination** [2024]
- **Qualified** for the **Regional Mathematics Olympiad (RMO)** [2023]
- **Merit Certificate** in **IOQM** for **four consecutive years** [2020–24]
- **AIR 37** at the **American Mathematics Competition** **(AMC 8)**, **MAA** – **1st in school** [2019]

SKILLS & INTERESTS

- **Programming Languages**: Python (primary), R; familiar with C++
- **Python Libraries**: NumPy, Pandas, PyTorch, Scikit-learn, LightGBM, PuLP
- **R Libraries**: dplyr, ggplot2, tidyr
- **Tools**: Git, GitHub, SQL, LaTeX
- **Interests**: Data Science, Statistical Modeling, Operations Research, Quantitative Finance
- **Spoken Languages**: English, Hindi, Bengali

EXPERIENCE

- **Future Focus Program** | **Optiver** [Jun 2026]
  - Completed Optiver's week-long summer trading and research program, covering market making, options and volatility and quantitative research through daily trading games, simulations and quant challenges.
  - Finished **2nd of 24 teams** in Optibook, the closing live-market competition, trading against bots and other participants for best P&L using aggressive market making and pairs trading.

PROJECTS

- **Comparative Study of BVP Solvers: PINN vs FDM vs Analytical** | **Prof. Ramij Rahaman** [Feb–Apr 2026]
  - Compared analytical, finite-difference (**FDM**), and physics-informed neural network (**PINN**) methods across linear, nonlinear (Bratu), and inverse boundary value problems.
  - Implemented FDM with Newton iteration for the nonlinear case and PINNs in `PyTorch`, using automatic differentiation, hard and soft boundary enforcement, and Adam + L-BFGS training.
  - **Recovered an unknown coefficient** from noisy observations by treating it as a trainable variable optimized jointly with the network, which FDM and analytical methods cannot do, and verified second-order FDM convergence.

- **Implied Volatility Surface Completion** | **NQFO 2026** [Apr 2026]
  - Built a two-layer ensemble (quadratic smile fit + **LightGBM**) to predict 1,699 missing implied volatility entries on a synthetic NIFTY50 options surface, anchoring the smile fit on same-day observed IVs to recast the task as within-date interpolation rather than cross-date prediction.

- Enforced arbitrage-free constraints: calendar-spread monotonicity via isotonic regression on total variance and butterfly positivity via a convexity constraint on the smile fit; left put-call parity unenforced after EDA confirmed genuine volatility skew rather than noise.
- Added **K-means** regime detection (calm/normal/turbulent) on ATM 1M IV and term-structure slope as a LightGBM feature, with moneyness-bucketed blend weights tuned by leave-one-out validation.

- **E-commerce Warehouse Location Optimization** | **Prof. Kaushik Jana** [Mar–May 2025]
  - Formulated warehouse placement as a facility location problem and solved it as a **MILP** in Python (`PuLP` with CBC solver) across India's top 10 e-commerce cities.
  - Modeled binary open/assign decisions under budget, warehouse-count, delivery-time, and single-assignment constraints, with Haversine great-circle distances driving the delivery-cost objective.
  - Selected **4 of 10** candidate cities as regional hubs (west/north/south/east) covering all demand centres within the budget, count, and delivery-time constraints, and validated solver optimality against the model formulation.

RELEVANT COURSES

- **Statistics & Probability:** Probability Theory, Statistical Inference, Regression & Linear Statistical Models, Nonparametric & Resampling Methods, Stochastic Processes, Monte Carlo & MCMC Methods
- **Data Science & Computing:** Data Structures & Algorithms, Database Management Systems, Data Analysis using R and Python
- **Mathematics:** Linear Algebra, Multivariate Calculus, Differential Equations, Real Analysis, Optimization & Numerical Methods, Graph Theory
- **Economics:** Microeconomics, Macroeconomics
- **Ongoing (Sem V):** Statistical Learning, Bayesian Inference, Advanced Statistical Methods II, Signal, Image & Text Processing

EXTRACURRICULAR ACTIVITIES

- Competitive Programming – **Codeforces** (Max Rating: 1449, 100+ solved problems)
- Enthusiast in Computer Systems, CPU/GPU Architecture and Hardware Optimization
- Speedsolving and Collecting a wide variety of Rubik's Cubes and Twisty Puzzles
- Recreational Poker player with interest in the game-theoretic and decision-making aspects of the game - **View Stats**